LivCast Capture Rev A — sparse symbols, pin names hidden, VIDEO_BUS simplified

SDI playback: CMS_HDMI → TMDS 1:2 → HDMI mux + ITE6021TN → VIDEO_BUS → GS2962A → BNC

VIDEO_BUS = 20-bit parallel PCMTA/DIN — expand as bus on PCB (not drawn pin-by-pin)

Power column left. Labels on wire stubs only. CMS split: U201–U205.

J101
Conn_01x02

GND
VIN_RAW

J102
Conn_01x04_MountingPin

GND
CC1_PWR
CC2_PWR
VBUS

U201
CMS_PCB5

USB_DN
USB_DP
GND
GND
GND
+3V3
+3V3
+3V3

U202
CMS_PCB6

HDMI_TX_CK_N
HDMI_TX_CK_P
HDMI_TX_D2_N
HDMI_TX_D2_P
HDMI_TX_D1_N
HDMI_TX_D1_P
HDMI_TX_D0_N
HDMI_TX_D0_P

U203
TPS62933DRLR

I2C_SCL
I2C_SDA
+3V3
GND
CC1_PWR
CC2_PWR
VBUS
GND
FB_SPI01
EPAD
EPAD

U204
TPS62933DRLR

SDA_SDO1
SCL
BOOT_01
SW_01
+3V3
GND

U205
CMS_PCB7

COL0_CK_N
COL0_CK_P
COL0_D3_N
COL0_D3_P
COL0_D2_N
COL0_D2_P
COL0_D1_N
COL0_D1_P
COL0_D0_N
COL0_D0_P

U206
TPS62933DRLR

SDA_TX
SCL_TX
BOOT_TX
SW_TX
+3V3
GND

U207
CMS_PCB8

COL1_CK_N
COL1_CK_P
COL1_D3_N
COL1_D3_P
COL1_D2_N
COL1_D2_P
COL1_D1_N
COL1_D1_P
COL1_D0_N
COL1_D0_P

U208
CMS_PCB9

SPI_CS
SPI_SCK
SPI_MISO
SPI_MOSI
I2C_SCL
I2C_SDA

J201
Conn_01x26

GND
GND
+12V_HDMI
HDMI_MON_HPD
HDMI_MON_DDC_SCL
HDMI_MON_DDC_SDA
HDMI_MON_CK_N
HDMI_MON_CK_P
HDMI_MON_D2_N
HDMI_MON_D2_P
HDMI_MON_D1_N
HDMI_MON_D1_P
HDMI_MON_D0_N
HDMI_MON_D0_P

U301
TMDS_Switch

GND
+3V3
TMDS_OE
HDMI_TX_CK_N
HDMI_TX_CK_P
HDMI_TX_D2_N
HDMI_TX_D2_P
HDMI_TX_D1_N
HDMI_TX_D1_P
HDMI_TX_D0_N
HDMI_TX_D0_P
HDMI_RX_CK_N
HDMI_RX_CK_P
HDMI_RX_D2_N
HDMI_RX_D2_P
HDMI_RX_D1_N
HDMI_RX_D1_P
HDMI_RX_D0_N
HDMI_RX_D0_P
HDMI_MON_CK_N
HDMI_MON_CK_P
HDMI_MON_D2_N
HDMI_MON_D2_P
HDMI_MON_D1_N
HDMI_MON_D1_P
HDMI_MON_D0_N
HDMI_MON_D0_P

U401
ITE6021TN

ITE6021_INTB
I2C_SCL
I2C_SDA
ITE6021_RESETB
+3V_HDMI
HDMI_RX_DDC_SCL
HDMI_RX_DDC_SDA
HDMI_RX_HPD
HDMI_RX_CK_N
HDMI_RX_CK_P
HDMI_RX_D2_N
HDMI_RX_D2_P
HDMI_RX_D1_N
HDMI_RX_D1_P
HDMI_RX_D0_N
HDMI_RX_D0_P
GND
GND
GND
+12V
+10V
+3V3
DE
VS
HS
PCLK
VIDEO_BUS

U501
GS2962A

SDI_TX_SPI_CS
SDI_TX_SPI_SCK
SDI_TX_SPI_MOSI
SMPTE_BYPASS
RATE_SEL0
RATE_SEL1
RST2I_L0
SL_STANDBY#
SDO_EN
GND
GND
GND
GND
+12V
+3V3
+3V3
+12V
SDI_TX_N
SDI_TX_P

J501
Conn_BNC_CES-TBS36-21

SDI_TX_P
GND

HDMI IN ingest → TBD (archive_sheets/03)

SDI IN ingest → TBD (archive_sheets/04, Ammicro)